

TransitGuard: An Edge-Processed Multi-Modal Data Logger

Team Spectre

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1. Project Overview & Problem Statement

Millions of dollars are lost annually due to the mishandling of fragile, high-value cargo during transit—ranging from sensitive electronics and semiconductor wafers to delicate prototyping equipment. Current market solutions, such as chemical "shock-watch" stickers, only measure a single variable and fail to record the specific timestamp of an event, rendering them ineffective for comprehensive auditing or fault attribution.

This proposal introduces **TransitGuard**, a compact, battery-powered smart logger. TransitGuard leverages edge-computing to track severe mechanical shocks, environmental extremes, and unauthorized access. By providing an immutable, timestamped audit trail of a package's journey, it offers a robust, commercially viable solution to a critical supply chain vulnerability.

2. How the MYOSA Board Will Be Used

The MYOSA motherboard will serve as the central processing unit and edge-computing gateway for the TransitGuard system. Its responsibilities are categorized as follows:

- **Interrupt-Driven Edge Processing:** Continuous data logging rapidly consumes memory and power. The MYOSA board will operate strictly on an interrupt basis, remaining in deep sleep and waking only to process and log specific "violation events" (e.g., when a shock exceeds a predefined G-force threshold).
- **Power Management Orchestration:** Utilizing the internal deep sleep capabilities of the onboard ESP32, the board will manage power states to ensure the logger can survive multi-day transits. It will utilize timer interrupts for periodic environmental checks and hardware interrupts for instant shock detection.
- **Data Aggregation & Sync:** Upon arrival at designated checkpoints, the board will leverage its integrated Wi-Fi capabilities to automatically push its timestamped audit log to a centralized cloud dashboard, ensuring the data is instantly accessible to stakeholders.
- **Internal Microcontroller Telemetry:** The ESP32's built-in Hall effect sensor will monitor for external magnetic hazards that could damage sensitive electronics, while the capacitive touch pins will act as proximity and anti-static breach detectors, maximizing telemetry at zero additional hardware cost.

3. Sensors and Actuators Used

TransitGuard achieves high-level "Sensor Fusion" by fully utilizing the MYOSA Mini IoT Kit modules to monitor a diverse array of failure modes:

- **MPU6050 (6-Axis IMU):** This sensor is critical for shock logging and orientation tracking. It will detect violent drops or impacts, recording the exact G-force. Furthermore, it will continuously track the Z-axis vector to verify if a package marked "This Way Up" was improperly stored upside down.
- **BMP180 (Barometric Pressure & Temperature):** This module monitors environmental extremes. It will ensure that sensitive cargo is not exposed to freezing or overheating conditions in cargo holds. Additionally, tracking pressure drops is vital for detecting rapid depressurization events in aircraft, which can destroy sealed electronic components.
- **APDS9960 (Ambient Light & Proximity Sensor):** Deployed for tamper detection. Placed inside a sealed, dark container, a sudden spike in ambient light readings will definitively log the exact timestamp of unauthorized access or package breach.
- **OLED Display (SSD1306):** This will act as the local readout interface. When the package reaches its destination, the receiver can view an immediate "health report" (e.g., Max

Shock, Max Temp, Tamper Status) directly on the package without needing external debugging tools.

4. Additional Circuitry and Software Requirements

To elevate this from a basic prototype to an industrial-grade solution, custom hardware interfacing and an optimized software architecture will be developed around the MYOSA board.

Custom Hardware & Circuits

- **Battery Management & Interface Shield:** The system will be powered by a 3.7V LiPo battery, requiring a custom power management circuit (incorporating a TP4056 module) for safe charging and regulation. This circuitry will be designed on a custom perfboard shield that mounts cleanly to the MYOSA board.
- **Hardware Protection:** To ensure the longevity of the microcontroller during field deployment, the external power inputs will integrate overvoltage and reverse-polarity protection. This will be achieved using carefully selected clamping components, specifically utilizing **1N5819 Schottky diodes** to safeguard the ESP32's sensitive power and logic pins.
- **External Data Storage:** An external SD Card module (communicating via SPI) will be integrated into the custom shield to allow for non-volatile storage of the audit log in a universally readable .CSV format.

Custom Software Architecture

- **Low-Level Interrupt Firmware:** The core firmware will be written to maximize battery life. The ESP32 will sleep while the MPU6050 remains active in a low-power mode. A dedicated hardware interrupt pin will connect the MPU6050 to the ESP32, triggering a wake-up routine the millisecond an impact threshold is breached.
- **Timestamping and Data Formatting (Python/C++):** Generating a valid audit trail requires precise timekeeping. The software will sync via NTP (Network Time Protocol) at the shipping origin. A software-based Real-Time Clock (RTC) routine will then maintain the time, timestamping every event. Algorithms will be written to filter noise from the raw IMU data, ensuring only verified impacts are logged.
- **Development Environment:** To maintain strict control over hardware-specific libraries and avoid version conflicts during the development of this complex sensor fusion logic, the software stack will be managed and deployed utilizing professional toolchains.